

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – V

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Class:TYIT	Batch:1
Date of Assignment: 21-07-2026	Date/Time of Submission: 28-07-2026

Part B - Program

- a) Create a web application to demonstrate the use of different types of Cookies.

CODE:-

```
using System;
using System.Web;
using System.Web.UI;
namespace Practical5c {
    public partial class WebForm1 : Page {
        protected void Page_Load(object sender, EventArgs e) {
            if(IsPostBack) {
                int ViewStateVal = Convert.ToInt32(ViewState["count"]) + 1;
                Label2.Text = "view state =" + ViewStateVal.ToString();
            } else { ViewState["count"] = 1; }
        }
        protected void Button2_Click(object sender, EventArgs e) {
            HttpCookie h = new HttpCookie("name");
            h.Value = TextBox1.Text;
            Response.Cookies.Add(h);
            Response.Redirect("Webform2.aspx");
        }
    }
}
Webform2
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
namespace Practical5c
{
    public partial class WebForm2 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (Request.Cookies["name"] != null)
            {
                Response.Write("Welcome" + Request.Cookies["name"].Value);
            }
        }
    }
}
```

OUTPUT:-

1

view state =2

View state

Omkar

Cookies

```
}
```

- b) Create a Product Table in SQL Server and perform Insert Operation using Entity Framework Core.

CODE:-

Migration

Add-Migration InitialCreate

Update-Database

Homecontroller.cs

using Microsoft.AspNetCore.Mvc;

using 5b..Models;

```
public class HomeController : Controller
```

```
{
```

```
    private readonly ApplicationDbContext _context;
```

```
    public HomeController(ApplicationDbContext context)
```

```
    {
```

```
        _context = context;
```

```
    }
```

```
    public IActionResult Index(){
```

```
        Product p = new Product();
```

```
        p.Name = "Laptop";
```

```
        p.Price = 50000;
```

```
        _context.Products.Add(p);
```

```
        _context.SaveChanges();
```

```
        return Content("Product Inserted Successfully");
```

```
    }
```

```
}
```

Product.cs

namespace 5b.Models

```
{
```

```
    public class Product
```

```
    {
```

```
        public int Id { get; set; }
```

```
        public string Name { get; set; }
```

```
        public int Price { get; set; }
```

```
    }
```

```
}
```

AppDbContext.cs

using Microsoft.EntityFrameworkCore;

namespace 5b..Models

```
{
```

```
    public class ApplicationDbContext : DbContext
```

```
    {
```

```
        public ApplicationDbContext(DbContextOptions<ApplicationDbContext> options)
```

```
            : base(options)
```

```
        {
```

```
        }
```

```
        public DbSet<Product> Products { get; set; }
```

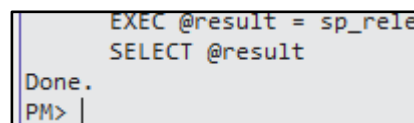
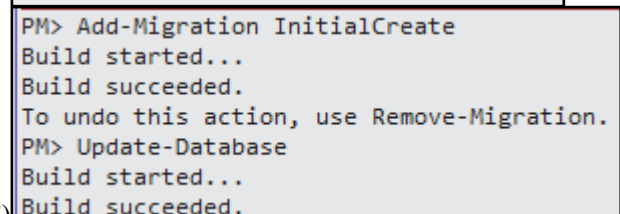
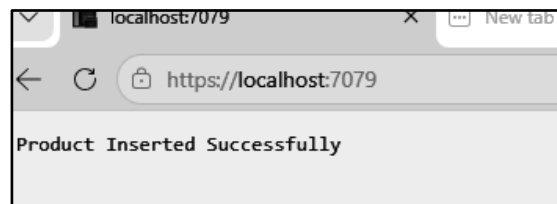
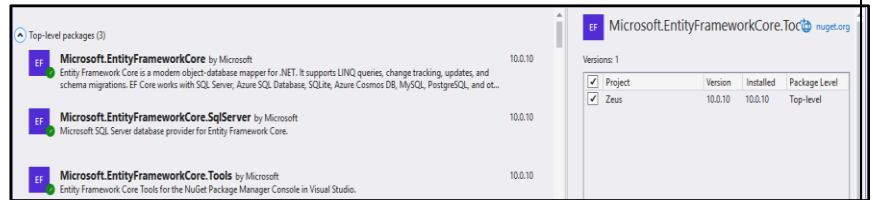
```
    }
```

```
}
```

appsetting.json

```
{
```

OUTPUT:-



```
"Logging": {
  "LogLevel": {
    "Default": "Information",
    "Microsoft.AspNetCore": "Warning"
  }
},
"AllowedHosts": "*",
"ConnectionStrings": {
  "DefaultConnection":
"Server=(localdb)\\MSSQLLocalDB;Database=ProductDB;Trusted_Connection=True;TrustServerCertificate=True;"
}
}
```

Program.cs

```
using Microsoft.EntityFrameworkCore;
using 5b.Models;
var builder = WebApplication.CreateBuilder(args);
builder.Services.AddControllersWithViews();
builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(
        builder.Configuration.GetConnectionString("DefaultConnection")));
var app = builder.Build();
app.UseStaticFiles();
app.UseRouting();
app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}");
app.Run();
```

- c) Develop a web application to perform CRUD Operations (Create, Read, Update, Delete) on the Employee table using EF Core.

CODE:-

Employee.cs

```
namespace P5_c.Models
{
    public class Employee
    {
        public string Department { get; set; }
        public int Id { get; set; }
        public string Name { get; set; }
        public double Salary { get; set; }}
}
```

AppDbContext.cs[New Folder in main proj named Data]

```
using Microsoft.EntityFrameworkCore;
using P5_c.Models;
namespace P5_c.Data
{
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options) : base(options) {}
        public DbSet<Employee> Employees { get; set; }}
}
```

appsettings.json

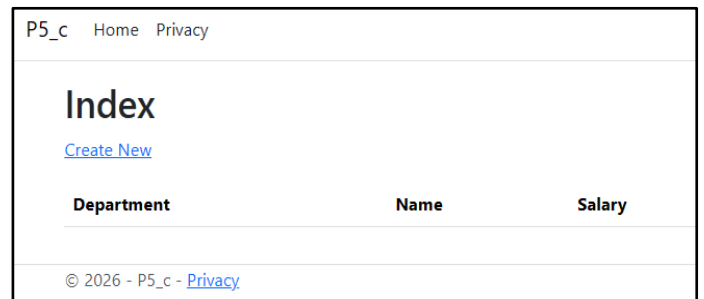
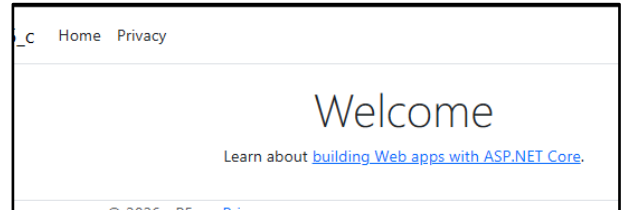
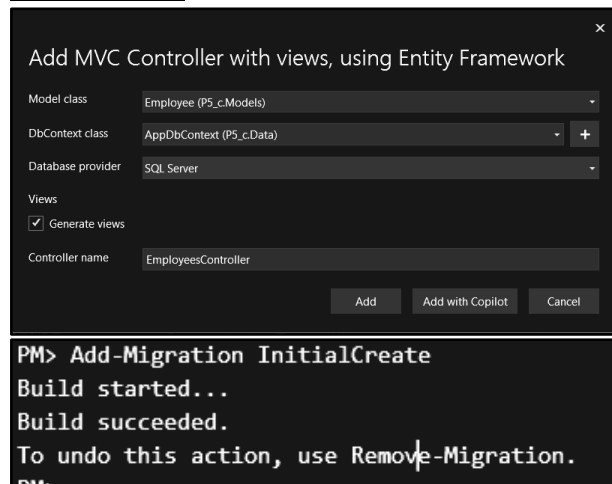
```
"AllowedHosts": "*",
```

```
"ConnectionStrings": {  
  "MyConnection":  
    "Server=(localdb)\\MSSQLLocalDB;Database=EmployeeDB;Trusted_Connection=True;TrustServerCertificate=True;"  
}
```

Program.cs

```
using Microsoft.EntityFrameworkCore;  
using P5_c.Data;  
builder.Services.AddDbContext<AppDbContext>(options =>  
    options.UseSqlServer(builder.Configuration.GetConnectionString("MyConnection")));
```

OUTPUT:-



Index

[Create New](#)

Department	Name	Salary	
Bsc.IT	Omkar Sanjay	1.00000000004E+19	Edit Details Delete

© 2026 - P5_c - [Privacy](#)

4) Authentication and Authorization in ASP.NET Core.

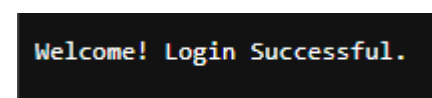
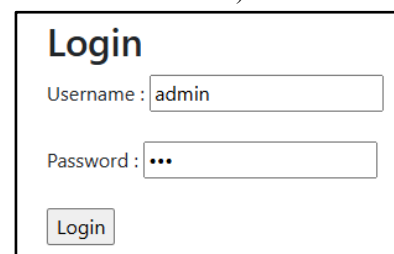
CODE:-

Program.cs

```
using Microsoft.AspNetCore.Authentication.Cookies;  
var builder = WebApplication.CreateBuilder(args);  
builder.Services.AddControllersWithViews();  
builder.Services.AddAuthentication(CookieAuthenticationDefaults.AuthenticationScheme)  
    .AddCookie(options =>  
    { options.LoginPath = "/Home/Index";  
    });  
builder.Services.AddAuthorization();  
var app = builder.Build();  
app.UseStaticFiles();  
app.UseRouting();  
app.UseAuthentication();  
app.UseAuthorization();  
app.MapControllerRoute(  
    name: "default",  
    pattern: "{controller=Home}/{action=Index}/{id?}");  
app.Run();
```

HomeController.cs

OUTPUT:-



```
using Microsoft.AspNetCore.Mvc;
using Microsoft.AspNetCore.Authorization;
using Microsoft.AspNetCore.Authentication;
using Microsoft.AspNetCore.Authentication.Cookies;
using System.Security.Claims;
namespace AuthDemo.Controllers
{
    public class HomeController : Controller
    {
        public IActionResult Index()
        {
            return View();
        }
        [HttpPost]
        public async Task<IActionResult> Index(string username, string password)
        {
            if (username == "admin" && password == "123")
            {
                var claims = new List<Claim>
                {
                    new Claim(ClaimTypes.Name, username)
                };
                var identity = new ClaimsIdentity(claims,
                    CookieAuthenticationDefaults.AuthenticationScheme);
                await HttpContext.SignInAsync(
                    CookieAuthenticationDefaults.AuthenticationScheme,
                    new ClaimsPrincipal(identity));
                return RedirectToAction("Secure");
            }
            ViewBag.Message = "Invalid Username or Password";
            return View();
        }
        [Authorize]
        public IActionResult Secure()
        {
            return Content("Welcome! Login Successful.");
        }
        public async Task<IActionResult> Logout()
        {
            await HttpContext.SignOutAsync(
                CookieAuthenticationDefaults.AuthenticationScheme);

            return RedirectToAction("Index");
        }
    }
}
```

Views/Home/Index.cshtml

```
<h2>Login</h2>
<form method="post"> Username :
    <input type="text" name="username" /><br /><br /> Password :
    <input type="password" name="password" /><br /><br />
    <input type="submit" value="Login" />
</form>
<p style="color:red">
    @ViewBag.Message
</p>
```